META-ANALYSIS

Individual Participant Data Meta-Analysis: Moderators of Response to Stimulants for Preschool Attention-Deficit/Hyperactivity Disorder

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Objective: This study aimed to identify moderators of efficacy of stimulants against placebo to inform personalized recommendations for treatment in preschool children (<6 years of age) with attention-deficit/hyperactivity disorder (ADHD).

Method: We acquired individual-level participant data from 2 randomized placebo-controlled trials (RCTs) of preschool ADHD: namely, MAPPA (8-week methylphenidate, 102 participants, Brazil) and SPD489-347 (6-week lisdexamfetamine, 148 participants, United States). We evaluated the moderator and predictor effects of baseline demographic (age, sex, race, ethnicity, maternal educational level) and baseline clinical (ADHD symptom severity, intelligence quotient, number of psychiatric comorbidities) characteristics, as available, on endpoint ADHD symptom severity scores. Data from each study were analyzed separately with linear mixed-effects model for repeated measures. For categorical variables, we also computed treatment effects (ie, stimulants vs placebo) within subgroups and, when possible, pooled them alongside subgroup data from the Preschoolers with Attention-Deficit/Hyperactivity Disorder Treatment Study (PATS) (5-week methylphenidate, 165 participants, United States) in random-effects meta-analyses.

Results: Stimulants had greater efficacy against placebo in White children compared to Black children when considering data from US studies. Older age was not a moderator of greater efficacy of stimulants against placebo, nor was it associated with worse ADHD symptom severity at endpoint. Greater baseline ADHD symptom severity was associated with higher ADHD symptom severity at endpoint independently of the assigned treatment group.

Conclusion: Race, but not older age or baseline ADHD symptom severity, may moderate the efficacy of stimulants for preschool ADHD. Given the post hoc nature of subgroup analyses, the findings should be interpreted as exploratory and viewed as hypotheses for confirmation in future studies.

Plain language summary: In this individual participant meta-analysis, the authors obtained data from 2 clinical trials of psychostimulants for preschool children (aged 6 years or younger) with attention-deficit/hyperactivity disorder (ADHD) to investigate associations between demographic, clinical characteristics, and treatment effects. One study was focused on methylphenidate and was conducted in Brazil with 102 participants. The other study focused on lisdexamfetamine and was conducted in the United States with 148 participants. Some analyses also included subgroup data from a study focused on methylphenidate and was conducted in the US with 165 participants. In the US studies, stimulants appeared to work better in White children than in Black children, while older age or clinical severity was not associated with effectiveness of stimulants. The authors emphasize that these findings are exploratory and should be confirmed in future studies before guiding clinical decisions.

Clinical trial registration information: Safety and Efficacy Study in Preschool Children Aged 4-5 Years With Attention-deficit/Hyperactivity Disorder (ADHD); https://clinicaltrials.gov/study/NCT03260205; Early Interventions in Children With Attention Deficit/Hyperactivity Disorder; https://clinicaltrials.gov/study/NCT02807870

Key words: randomized controlled trial; amphetamine; methylphenidate

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Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental disorder characterized by a pervasive, persistent, and developmentally inappropriate pattern of inattention and/or hyperactivity–impulsivity that interferes with development and/or functioning.[1] ADHD symptoms usually emerge in the first years of life,[2] and a diagnosis may be possible in children as young as 3 to 4 years.[3] Population-based studies from the United States and Scandinavian countries estimate that around 2 in 100 preschool children (ie, <6 years of age) may receive a diagnosis of ADHD.[4-6] Preschool ADHD has been associated with interference in social and academic development,[7] substantial economic burden,[8] and possibly increased risk of mortality attributed to physical injuries.[9]

Treatment could reduce the impact of ADHD early in life. A large body of evidence from randomized controlled trials (RCTs) has shown that stimulants (ie, methylphenidate and amphetamines) lead to large reductions in the

severity of ADHD symptoms in school-aged children and adolescents.[10,11] However, less robust evidence base specifically for preschool children is available to guide practitioners and guideline developers.[12] A systematic review with meta-analysis could find only 5 RCTs with a median treatment duration of 4 weeks and approximately 500 participants testing stimulants for this patient group.[13] Those RCTs showed that stimulants, compared against placebo, had only a moderate effect size on the reduction in the severity of ADHD symptoms. In addition, although stimulants have mostly been associated with higher rates of mild adverse events only,[14] longer-term RCTs in the preschool population are lacking, prompting concerns around the safety of those medications, in particular regarding the cardiovascular system, somatic growth, and brain development.[15-17] In this context, clinical practice guidelines (CPGs) more conservatively suggest that stimulants may be an option for preschoolers who continue to experience interference in functioning after a trial of a nonpharmacological intervention.[18,19] Nevertheless, there is concern that, unintentionally, these recommendations may contribute to the delayed initiation of stimulants and the overuse of non–evidence-based interventions, such as use of antipsychotics, for preschoolers in the community.[20-22]

Practitioners may be more inclined to prescribe stimulants for those preschool children who are more likely to experience greater benefits with the medication. Additional insights on stimulant treatment in this population are needed to inform such personalized recommendations. Specifically, published analyses of RCTs consider average effects (effects are valid at the group level), but there is substantial individual-level variability, and some preschoolers may experience greater benefits with stimulants than the average.[23] Demographic and clinical characteristics that are routinely collected at patient intake offer an opportunity to attempt to identify those individuals who may be more likely to benefit from stimulants. However, to date, only one previous study has examined this issue using data from the initial 5-week phase of the landmark National Institute of Mental Health (NIMH)–funded Preschoolers with Attention-Deficit/Hyperactivity Disorder Treatment Study (PATS).[24] This study reported that the number of comorbid disorders was inversely associated with the efficacy of methylphenidate against placebo in terms of reductions of ADHD symptom severity.[25] Nonetheless, post hoc secondary analyses of RCTs may be particularly prone to both type I and type II errors.[26-28] Therefore, it is critical that findings from moderator analyses based on one RCT be replicated in other samples and show robustness in pooled analyses of studies.[28] Notably, in more recent years, other RCTs of stimulants in preschoolers have been conducted, offering the opportunity to test replicability of the finding from PATS in other samples, as well as to identify other moderators of response to stimulants in preschoolers.

To address this critical gap, we conducted, for the first time, moderation analyses using individual participant data from 2 RCTs of stimulants vs placebo for preschoolers with ADHD and pooled findings across studies when possible.

METHOD

Study Design

The current study builds upon a previous systematic review of placebo-controlled RCTs of stimulants for preschool ADHD. In that previously published study, which was preregistered with PROSPERO CRD42022348597, literature searches on 4 online databases (Cochrane Central Register of Controlled Trials; Embase; PubMed; WHO International Clinical Trials Registry Platform) were carried out up to July 2022 to identify 5 double-blinded RCTs of stimulants vs placebo including 489 preschoolers with a primary diagnosis of ADHD according to the *DSM* criteria.[13]

In this study, we sought to acquire individual-level participant data (IPD) from the 5 RCTs[24,29-32] that were included in the prior study. By contacting study authors or sponsors, we acquired IPD from 2 RCTs: (1) MAPPA, an 8-week RCT of methylphenidate conducted in a specialist center in São Paulo, Brazil, with participants enrolled from August 2016 to October 2019 (NCT02807870)[30]; and (2) SPD489-347, a 6-week RCT of lisdexamfetamine conducted in 44 centers in the United States between September 2017 and October 2018 (NCT03260205).[31] Additional details on each study are available in their corresponding published reports[30,31] and ClinicalTrials.gov records, all of which are openly available.

We were not able to include IPD from 2 RCTs, as investigators no longer had access to the dataset[32] or did not respond to our inquiry.[29] Also, we were not able to include IPD from PATS because we were unable to identify the variable corresponding to the randomized groups in the dataset from NIMH Data Archive. In fact, we contacted the NIMH Data Archive Help Desk, 6 investigators from PATS, and the data management center for the study, but we were unable to retrieve any other information. This was understandable, as the study was conducted 20 years ago. However, the investigators from PATS had previously conducted moderator analyses and published the group-level statistics (ie, mean, SD) of ADHD symptom severity with methylphenidate or placebo at endpoint for each subgroup of candidate categorical moderators.[25] Therefore, we were able to extract those data and to calculate within-subgroup effect sizes for pooling in meta-analysis alongside information from MAPPA and SPD489-347.

Participants

According to the eligibility criteria of the RCTs included in our study, children had to be 5.5 years of age or younger and to have a primary diagnosis of ADHD according to *DSM* criteria (edition IV onward) with moderate or severe baseline core ADHD symptoms.

Children were required to attend school or daycare with same-aged peers and to have an intelligence quotient (IQ) of ≥70 in MAPPA and PATS, or a score of ≥70 on the Peabody Picture Vocabulary in SPD489-347.

Children with co-occurring autism or affective or psychotic disorders were not eligible for participation. In addition, in SPD489-347 and PATS, children were eligible only if they had failed a previous adequate trial of a nonpharmacological intervention (in SPD489-347, investigator defined based on patient history; in PATS, parent training administered by PATS investigators) unless their symptoms were judged to be severe enough to skip that step.

PATS and SPD489-347 also excluded participants who had evidence of nonresponse or poor tolerability to stimulants, although all but 2 participants from MAPPA were stimulant naive.

Interventions

In MAPPA, methylphenidate was titrated from 0.3 mg/kg/d to 0.7 mg/kg/d in the first 4 weeks and then up to 1.25 mg/kg/d between the 4th and 8th weeks if there was room for improvement and no limiting adverse events (daily dose mean [SD] = 27 [6] mg/d). It was prescribed adjuvant to an educational intervention that focused on typical development and had no active component addressing ADHD symptoms.

In SPD489-347, lisdexamfetamine was randomly prescribed at fixed-doses of 5 mg/d, 10 mg/d, 20 mg/d, or 30 mg/d with forced weekly dose escalation. If the assigned dose was not tolerated, participants were required to discontinue treatment. We considered data only from groups assigned to doses of 10 mg/d through 30 mg/d following the primary efficacy analysis for the RCT, in line with the protocol established a priori.[31]

In PATS, methylphenidate was prescribed at multiple fixed doses (3.75, 7.5, 15, 22.5 mg/d) for 1 week each. We considered data only for the dose of 22.5 mg/d, in accordance with the previous study that reported moderation analyses in PATS.[25]

Outcome

The outcome of this study was ADHD symptom severity at endpoint, without any restrictions on rating scale or informant. When multiple rating scales or informants were available, we chose the study-defined primary outcome measure. Additional details on the rating scales that were used can be found in Supplement 1, available online.

From MAPPA, we used the combined parent and teacher Swanson, Nolan and Pelham–IV (SNAP-IV) scores,[33] measured at baseline, week 4, and week 8 (endpoint).

From SPD489-347, we considered the clinician ADHD Rating Scale–IV–Preschool (ADHD-RS-IV-PS) scores,[34] measured weekly from baseline through week 6 (endpoint).

From PATS, we used the combined parent and teacher Conners, Loney and Milich Rating Scale (CLAM) and Swanson, Conners, Milich and Pelham Scale (SKAMP) scores,[35] measured just once at the end of the week.

Putative Moderators and Predictors

Candidate variables were selected considering current recommendations from CPGs, potential to inform clinical practice, and data availability across at least 2 RCTs to enable replicability and pooling of data in meta-analyses. We selected both baseline demographic (age, sex, race, ethnicity, maternal educational level) and baseline clinical (baseline ADHD symptom severity, IQ, and number of co-occurring psychiatric conditions) characteristics.

Age and ADHD symptom severity were chosen because some CPGs make differential recommendations regarding stimulant treatment for preschool- vs school-aged children and for individuals with mild-to-moderate vs severe ADHD symptoms.[19,36] Both variables were analyzed continuously. However, for baseline age, we also considered a dichotomized variable (<5 years; ≥5 years) to reflect the National Institute for Health and Care Excellence (NICE) differential treatment recommendations for children under 5 years of age and those 5 years and over.[19] Sex (male participants, female participants), race (Black, White), and ethnicity (Hispanic, non-Hispanic) were analyzed, considering that minorities are underrepresented in ADHD RCTs and clinical research,[37,38] and more research is needed to characterize whether there is differential treatment response based on sex, race, and ethnicity. The number of baseline co-occurring psychiatric conditions (continuous and categorized into none, 1 or 2, and 3 or more) was included because it was the only significant finding from the previous study that reported moderation analyses in PATS.[25] Similarly, we also included maternal educational level (dichotomized into high school vs post–high school education) and children's IQ (continuous and dichotomized into ≤85 or >85) following the previous study with PATS.[25]

Statistical Analyses

To analyze IPD from MAPPA and SPD489-347, we initially assessed the distribution of baseline demographic and clinical characteristics of participants across arms and

then reproduced the primary efficacy analysis as described in the published reports.[30,31] Analyses adhered to the intention-to-treat principle and adopted a linear mixed-effects model for repeated measures (MMRM) using data from all available timepoints and an interaction term visit*group to assess the effect of stimulant vs placebo at endpoint (ie, time treated as a categorical variable). Additional details about the statistical models from each study, as well as covariates that were included in each analysis, are available from their corresponding published reports.[30,31] We compared our results with those from the published reports to ensure that there were no discrepancies.

We considered that baseline characteristics were a moderator if the effect size of stimulants against placebo on ADHD symptom severity at endpoint was not the same for all values of the candidate variable.[23,39] To assess the moderator effect of putative variables, we added a 3-way interaction term variable*visit*treatment to the MMRM. We considered that baseline characteristics were a predictor if there was a main effect on ADHD symptom severity, but the comparative effect size between interventions was the same for all values of the candidate variable.[23,39] To assess the predictor effect of putative variables, we considered the corresponding main effect coefficient from the model including the 3-way interaction term. We used a type III analysis of variance with the Satterthwaite method to probe statistical significance of putative variables. Candidate variables were tested individually, one at a time.

To pool data from MAPPA and SPD489-347 in meta-analyses, we extracted the standardized regression coefficients (β) and their corresponding standard errors (SE), estimated by using standardized ADHD symptom severity as the outcome. We used the within-study baseline standard deviation across arms for standardization. In addition, to enable pooling within subgroup effect sizes of methylphenidate vs placebo from PATS, for categorical variables we also calculated the within-subgroup effect sizes of methylphenidate and lisdexamfetamine vs placebo at endpoint for MAPPA and SPD489-347, respectively, using the endpoint standard deviation across arms for standardization. We then pooled those data to compute summary standardized mean difference (SMD) of stimulant against placebo within each subgroup in random-effects meta-analyses weighted by the inverse of the variance. We then ran subgroup tests to probe differences in stimulant efficacy across subgroups.

Analyses were conducted in R (version 4.4.2) with packages "*lme4*" (version 1.1-35.5)[40] and "*meta*" (version 8.0-1).[41] Statistical significance was determined at the level of $p < .05$.

RESULTS

In MAPPA, 102 children were randomized to methylphenidate (n = 51) or placebo (n = 51). Of those, 3 participants (1 in methylphenidate, 2 in placebo) discontinued treatment. All 102 participants were included in the analyses. In SPD489-347, 159 children were randomized to lisdexamfetamine (n = 113) or placebo (n = 46). Of those, 41 participants (32 in lisdexamfetamine, 9 in placebo) discontinued treatment. All randomly assigned participants receiving at least 1 dose of lisdexamfetamine and having at least 1 postbaseline assessment (148 of 159, 93%) were included in the analyses. Table 1 lists the baseline demographic and clinical characteristics that were available from each study, stratified by treatment group. In PATS, 165 children enrolled in the crossover titration phase,[24] and the distribution of baseline characteristics has been previously reported.[25]

Baseline Demographic Characteristics

Age. Age did not moderate response to stimulants in either MAPPA (β = –0.02 for every year from 3 years of age, 95% CI = –0.66 to 0.62, $F = 0.91$, $p = .41$) or SPD489-347 (β = 1.02 for every year from 3 years of age, 95% CI = –0.05 to 2.04, $F = 1.28$, $p = .26$). Meta-analysis across the 2 studies similarly did not indicate any differential association between baseline age and ADHD symptom severity at endpoint across treatment arms ($\beta = 0.41$ for every year from 3 years, 95% CI = –0.59 to 1.41, $p = .42$).

When considering subgroups based on a cutoff of 5 years, baseline age similarly did not moderate response to stimulant in either MAPPA or SPD489-347 (Table 2). Meta-analysis with all 3 studies also did not indicate any significant subgroup differences between children under 5 years of age and those 5 years or older ($Q = 0.13$, $p = .72$) (Figure 1; Figure S1, available online).

Finally, when analyzed either as a continuous or a categorical variable, baseline age was not a significant predictor of ADHD symptom severity at endpoint in either study (Table 3) or in meta-analysis ($\beta = 0.26$ for every year from 3 years, 95% CI = –0.21 to 0.73, $p = .27$).

Sex. Sex did not significantly moderate response to stimulant in either MAPPA or SPD489-347 (Table 2). Meta-analysis with all 3 studies did not indicate subgroup differences between female and male participants ($Q = 1.26$, $p = .26$) (Figure 1; Figure S2, available online).

Sex was also not a predictor of ADHD symptom severity at endpoint in either study (Table 3) or in meta-analysis (β = –0.10 for female participants compared to male participants, 95% CI = –0.73 to 0.52, $p = .75$).

TABLE 1 Baseline Characteristics of Participants From MAPPA and SPD489-347 by Treatment Group

	MAPPA				SPD489-347			
	Methylphenidate		Placebo		Lisdexamfetamine		Placebo	
	n / Mean	% / SD	n / Mean	% / SD	n / Mean	% / SD	n / Mean	% / SD
Age, y (continuous)	5	0.66	5.04	0.58	5.04	0.56	5.1	0.53
Age, y (categorical)								
≥5 y	29	56.9	29	56.9	61	59.2	25	55.6
<5 y	22	43.1	22	43.1	42	40.8	20	44.4
Sex								
Male participants	45	88.2	41	80.4	70	68.0	29	64.4
Female participants	6	11.8	10	19.6	33	32.0	16	35.6
Race								
White	—	—	—	—	51	49.5	19	42.2
Black	—	—	—	—	44	42.7	23	51.1
Ethnicity								
Non-Hispanic	—	—	—	—	86	83.5	37	82.2
Hispanic	—	—	—	—	17	16.5	8	17.8
Maternal educational level								
High school	36	70.6	21	41.2	—	—	—	—
Post high school	15	29.4	30	58.8	—	—	—	—
ADHD symptom severity (continuous)	37.38	7.76	35.92	6.60	43.28	6.45	44.33	6.42
IQ (continuous)	88.45	11.16	90.59	11.79	—	—	—	—
IQ (categorical)								
≤85	22	43.1	19	37.3	—	—	—	—
>85	29	56.9	32	62.7	—	—	—	—
Psychiatric comorbidities (continuous)	1.21	1.14	0.78	0.85	—	—	—	—
Psychiatric comorbidities (categorical)								
None	15	29.4	23	45.1	—	—	—	—
1 or 2	29	56.9	26	51	—	—	—	—
≥3	7	13.7	2	3.9	—	—	—	—

Note: For continuous variables, values are means and SDs. For categorical variables, values are counts (n) and percentages (%). ADHD = attention-deficit/hyperactivity disorder; IQ = intelligence quotient.

Race. In SPD489-347, 67 (45%) and 70 (47%) of the 148 individuals included in the analyses self-identified as Black and White, respectively, with the remaining 11 individuals scattered across different categories (1 [0.7%] American Indian, Alaskan Native; 5 [3%] multiple; 4 [3%] "other"; 1 [0.7%] missing). Modeling of race as Black vs White, with the remaining individuals entered in the model as a third category, indicated that race did not moderate response to stimulant in SPD489-347 (Table 2). However, meta-analysis with PATS indicated significant subgroup differences ($Q = 7.13$, $p = .008$), with large significant effects of stimulants against placebo on White (SMD = –0.87, 95% CI = –1.18 to –0.56, $p < .0001$) compared to medium-to-small non-significant effects on Black (SMD = –0.28, 95% CI = –0.58 to 0.03, $p = .08$) (Figure 1; Figure S3, available online).

There was no evidence that race was a predictor of ADHD symptom severity at endpoint in SPD489-347 (Table 3).

Ethnicity. In SPD489-347 there were 25 (17%) Hispanic participants, and modeling of Hispanic vs non-Hispanic participants indicated that ethnicity did not moderate response to stimulant in SPD489-347 (Table 2). Meta-analysis with PATS did not indicate subgroup differences between Hispanic and non-Hispanic participants ($Q = 1.62$, $p = .20$) (Figure 1; Figure S4, available online).

There was no evidence that ethnicity was a predictor of ADHD symptom severity at endpoint in SPD489-347 (Table 3).

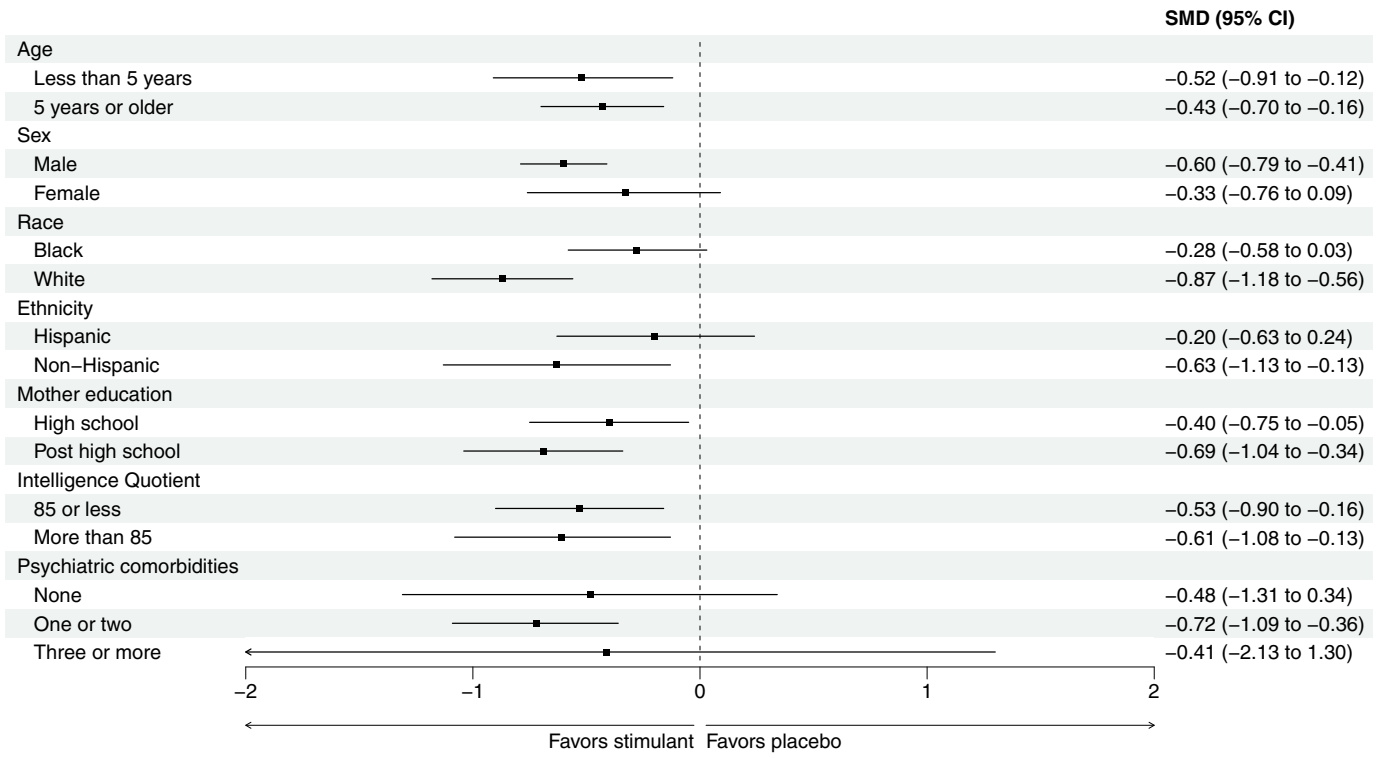
Maternal Educational Level. Maternal educational level did not significantly moderate response to stimulant in

TABLE 2 Attention-Deficit/Hyperactivity Disorder (ADHD) Symptom Severity Sum Scores by Treatment Group Across Subgroups for MAPPA and SPD489-347

	MAPPA						SPD489-347					
	Methylphenidate		Placebo		Moderation		Lisdexamfetamine		Placebo		Moderation	
	Baseline	Week 8	Baseline	Week 8			Baseline	Week 6	Baseline	Week 6		
	LSM (SE)	LSM (SE)	LSM (SE)	LSM (SE)	*F*	*p*	LSM (SE)	LSM (SE)	LSM (SE)	LSM (SE)	*F*	*p*
Age, y					0.45	.64					0.96	.45
≥5	36.1 (2.0)	27.2 (2.0)	36.8 (2.0)	33.6 (2.0)			43.9 (1.4)	30.6 (1.5)	44 (2.2)	33.9 (2.2)		
<5	37.9 (2.1)	28.9 (2.1)	34.6 (2.1)	29.5 (2.1)			43.8 (1.7)	29.4 (1.8)	43.5 (2.5)	37.1 (2.5)		
Sex					0.28	.76					1.45	.19
Male participants	36.8 (1.8)	27.9 (1.8)	36.1 (1.9)	32.7 (1.9)			43.8 (1.3)	29.6 (1.4)	43.8 (2.1)	36.6 (2.1)		
Female participants	37 (3.6)	28.0 (3.6)	34.8 (2.9)	28.6 (2.9)			44.0 (1.9)	31.4 (2.0)	43.9 (2.7)	33.1 (2.7)		
Race					—	—					0.78	.67
White	—	—	—	—			44.3 (1.6)	28.5 (1.6)	44.3 (2.5)	38.1 (2.5)		
Black	—	—	—	—			43.5 (1.7)	31.4 (1.7)	43.6 (2.3)	34.2 (2.3)		
Ethnicity					—	—					0.68	.67
Non-Hispanic	—	—	—	—			43.9 (1.3)	29.5 (1.3)	43.9 (1.9)	34.7 (1.9)		
Hispanic	—	—	—	—			43.7 (2.7)	33.8 (2.8)	43.6 (3.9)	38.8 (4.2)		
Maternal educational level					0.14	.87					—	—
High school	37.3 (1.8)	28.4 (1.8)	37.3 (1.9)	32.7 (1.9)			—	—	—	—		
Post high school	35.7 (2.5)	26.8 (2.5)	33.7 (2.1)	30.5 (2.2)			—	—	—	—		
IQ					1.09	.34					—	—
≤85	38.0 (2.1)	27.6 (2.1)	36.3 (2.2)	32.9 (2.3)			—	—	—	—		
>85	35.9 (2.0)	28.2 (2.0)	35.5 (1.9)	31.3 (1.9)			—	—	—	—		
Psychiatric comorbidities					0.98	.42					—	—
None	36.4 (2.4)	29.0 (2.4)	35.1 (2.1)	29.5 (2.1)			—	—	—	—		
1 or 2	37.3 (1.9)	29.1 (1.9)	36.2 (2.0)	33.7 (2.0)			—	—	—	—		
≥3	38.0 (3.3)	21.8 (3.5)	41.1 (6.0)	34.9 (6.0)			—	—	—	—		

***Note**: Values represent least square means (LSM) on the original instrument scale (MAPPA: SNAP-IV; SPD489-347: ADHD-RS-IV-PS) with standard errors (SE) from linear mixed-effects models for repeated measures including data from all available timepoints.* F *values and* p *values were calculated based on type III analysis of variance for the 3-way interaction term. IQ = intelligence quotient.*

FIGURE 1 Forest Plot of the Comparative Effects of Stimulant Against Placebo on Endpoint Attention-Deficit/Hyperactivity Disorder (ADHD) Symptom Severity in Subgroups of Preschool Children With ADHD



Note: *The figure illustrates within-subgroups standardized mean differences (SMD) alongside 95% confidence intervals.*

MAPPA (Table 2). Meta-analysis with PATS did not indicate subgroup differences between children whose mothers had a high school degree and those whose mothers had a post–high school degree ($Q = 1.29$, $p = .26$) (Figure 1; Figure S5, available online).

Maternal educational level was also not a predictor of ADHD symptom severity at endpoint in MAPPA (Table 3).

Baseline Clinical Characteristics

Baseline ADHD Symptom Severity. Baseline ADHD symptom severity did not moderate response to stimulant in either MAPPA ($\beta = –0.03$, 95% CI = –0.09 to 0.02, $F = 1.48$, $p = .23$) or SPD489-347 ($\beta = 0.06$, 95% CI = –0.03 to 0.14, $F = 0.44$, $p = .85$). Meta-analysis across the 2 studies similarly did not indicate a differential association between baseline ADHD symptom severity and ADHD symptom severity at endpoint across treatment arms ($\beta = 0$, 95% CI = –0.08 to 0.09, $p = .92$).

However, baseline ADHD symptom severity was a predictor of ADHD symptom severity at endpoint in both MAPPA and SPD499-347 (Table 3), with meta-analysis indicating an increase of 0.14 SD (95% CI = 0.11 to 0.17) in ADHD symptom severity at endpoint for every 1-point increase in baseline ADHD symptom severity.

Intelligence Quotient. Intelligence quotient (IQ) did not moderate response to stimulant in MAPPA ($\beta = 0.01$, 95% CI = –0.02 to 0.04, $F = 0.22$, $p = .80$). When considering subgroups based on a cutoff of 85, IQ similarly did not moderate response to treatment in MAPPA (Table 2). Meta-analysis with PATS also did not indicate subgroup differences between children with an IQ of 85 or under and those with a higher IQ ($Q = 0.07$, $p = .80$) (Figure 1; Figure S6, available online).

IQ was also not a predictor of ADHD symptom severity at endpoint in MAPPA (Table 3).

Psychiatric Comorbidities. The number of baseline psychiatric comorbidities moderated response to stimulant in MAPPA; however, unlike in PATS, a higher number was associated with larger treatment effects of methylphenidate against placebo at endpoint ($\beta = –0.50$, 95% CI = –0.91 to –0.1, $F = 3.0$, $p = .05$). When considering subgroups based on cutoffs of none, 1 or 2, and 3 or more baseline psychiatric comorbidities, there was no evidence of

TABLE 3 Main Effects of Candidate Predictors on Attention-Deficit/Hyperactivity Disorder (ADHD) Symptom Severity at Endpoint

	MAPPA				SPD489-347			
	β	95% CI	*F*	*p*	β	95% CI	*F*	*p*
Age[a] (continuous)	0.33	−0.2 to 0.88	0	0.99	0.05	−0.86 to 0.96	0.54	0.46
Age, y (Ref: ≥5 y)			0.22	0.64			0.03	0.87
<5 y	−0.3	−0.94 to 0.33			−0.07	−1.03 to 0.9		
Sex (ref: male participants)			0	0.92			0	0.95
Female participants	−0.18	−0.98 to 0.62			0.01	−0.98 to 1.0		
Race (ref: White)			—	—			0.14	0.70
Black	—	—			−0.01	−1.07 to 0.91		
Ethnicity (ref: Non-Hispanic)			—	—			0.07	0.79
Hispanic	—	—			0.04	−1.24 to 1.32		
Maternal educational level (ref: high school)			1.16	0.28			—	—
Post high school	−0.50	−1.14 to 0.15			—	—		
ADHD symptom severity	**0.14**	**0.10 to 0.17**	**191**	**< 0.001**	**0.15**	**0.08 to 0.23**	**23.4**	**< 0.0001**
IQ (continuous)	0	−0.02 to 0.03	0	0.95	—	—	—	—
IQ (ref: >85)			0.07	0.79			—	—
≤85	0.11	−0.54 to 0.76			—	—		
Psychiatric comorbidities (continuous)	0.22	−0.26 to 0.71	0.65	0.42	—	—	—	—
Psychiatric comorbidities (ref: none)			0.53	0.58			—	—
1 or 2	0.15	−0.5 to 0.79			—	—		
≥3	0.84	−0.83 to 2.5			—	—		

Note: Boldface type indicates statistical significance. Values represent standardized regression coefficients (β) alongside 95% confidence intervals for the main effect of the variable, one at a time. F values and p values were calculated based on type III analysis of variance. Statistical significance indicates that the variable is a non-specific predictor of ADHD symptom severity at endpoint irrespective of treatment group (MAPPA: methylphenidate or placebo; SPD489-347: lisdexamfetamine or placebo). ADHD = attention-deficit/hyperactivity disorder; IQ = intelligence quotient; ref = reference.
[a]For every year starting from 3 years.

moderation (Table 2). Similarly, meta-analysis with PATS did not indicate subgroup differences between children with none, 1 or 2, and 3 or more baseline psychiatric comorbidities ($Q = 0.36$, $p = .84$) (Figure 1; Figure S7, available online).

The number of psychiatric comorbidities, when analyzed as either a continuous or categorical variable, was not a predictor of ADHD symptom severity at endpoint in MAPPA (Table 3).

DISCUSSION

In this study, we aimed to generate an evidence base that could inform personalized recommendations of treatment for preschoolers with ADHD. For the first time, we analyzed individual-level participant data from 2 placebo-controlled RCTs of stimulants (lisdexamfetamine, methylphenidate) for preschool ADHD.[30,31] When possible, we pooled the data from those studies together alongside subgroup-level treatment effects from PATS,[24] extracted from a previous publication reporting on moderator analyses conducted by the PATS investigators.[25] Our findings indicated that race may moderate stimulant efficacy, as White children experienced a larger treatment effect compared to Black children. There were no other variables with a significant moderating effect, including age or baseline ADHD symptom severity. Finally, our results indicated that only baseline ADHD symptom severity was a non-specific predictor, with higher baseline ADHD symptom severity being associated with higher ADHD symptom severity at endpoint independently of the assigned treatment group. Taken together, those findings, which are exploratory given the nature of post hoc subgroup analyses, generate hypotheses that should be examined in future empirical studies.

We found smaller effects of stimulants against placebo in Black children compared to White children through pooled subgroup analyses of data from SPD489-347 and PATS. To the best of our knowledge, our study is the first to report such finding for preschoolers with ADHD. Notably, a sizable proportion of ADHD RCTs do not account for race,[37,42] and the associations between race and treatment outcomes with stimulants may have been

overlooked thus far. There are several possible explanations for this finding. Given that race is a social construct, a thorough consideration of factors related to structural racism is important. For example, the diminished efficacy of stimulants on Black vs White preschool children with ADHD could be confounded by larger exposure of minoritized populations to socioeconomic hardship (eg, as shown in the Multimodal Treatment of Attention Deficit Hyperactivity Disorder [MTA] study).[43] However, careful consideration about confounding is warranted, as some factors may instead be mediators and may underlie the pathway between race and diminished stimulant efficacy. For instance, previous research has shown that Black individuals may have lower rates of medication use and compliance compared to their White counterparts.[44,45] In turn, decreased adherence may underlie the association between race and stimulant efficacy. The distinction between confounders and mediators is relevant because interventions focused on the latter may contribute to improved outcomes, and the field should focus on this topic to promote equity around stimulant treatment. In addition, future studies should also examine the possible impact of teacher or clinician rater biases on the association between Black children and reduced stimulant efficacy.[46,47]

Notably, we did not find any moderating effect of baseline age on stimulant efficacy through pooled subgroup analyses of data from the 3 RCTs. Traditionally, clinical lore considers that stimulants are less efficacious in preschool- than in school-aged children; however, there are no substantiative data to support this claim. Most often, practitioners point to differences in the magnitude of effect sizes of stimulants against placebo across age groups, but comparative evidence across RCTs is limited because of potential residual confounding. For instance, stimulants have been prescribed at relatively low doses in RCTs of preschoolers, but flexible titration until maximum licensed doses has been associated with greater efficacy in RCTs of other pediatric populations.[48] In addition, RCTs of preschoolers typically require a previous trial of a non-pharmacological intervention,[24,29,31] which in turn may contribute to overrepresentation of severe cases that may have worse treatment outcomes. Our findings showed that, within the preschool population, older age was not associated with greater stimulant efficacy compared to placebo. Whether our findings indicate that recommendations from CPGs should separate preschoolers from school-aged children at 6 years as the age cut-off or should simply abolish differences across age groups remains to be established.

In addition, we did not find a moderating effect of baseline ADHD symptom severity on stimulant efficacy. Some CPGs (eg, German and Dutch guidelines for ADHD) indicate differential recommendations regarding stimulants for mild-to-moderate vs severe ADHD. Understandably, practitioners may be more inclined to prescribe stimulants in scenarios where ADHD symptoms are more severe to maximize benefits against risks. There is no empirical non-arbitrary ADHD symptom severity threshold regarding impairment,[49] and some individuals with low ADHD symptom severity may still be considered "severe" based on clinical judgment that also accounts for ADHD-related impairment. Therefore, although our findings are not necessarily contrary to current recommendations, they reinforce that assessment of "severe" ADHD should not be based solely on a threshold score in ADHD symptom rating scales, at least in preschoolers.

Importantly, although subgroup analyses are appealing, we stress that the overall finding from an RCT usually represents the best estimate of the treatment effect compared with those reported in post hoc subgroup analyses.[50,51] For instance, given the limited number of Black individuals in the RCTs, it remains to be determined whether the lower effect of stimulants vs placebo found in our study is representative of the true comparative effect in the population. In addition, there is risk that the association between race and lower stimulant efficacy represents a spurious significant finding purely by chance, arising from multiple comparisons. At the same time, the absence of differences in stimulant efficacy across age subgroups could merely reflect insufficient statistical power. Altogether, the findings from our study should be interpreted as generating additional hypotheses, which should be directly tested by experimental research in the future. Given that moderation analyses have the potential to inform treatment selection,[52] it would be particularly helpful if the field attempted to confirm the findings of (lack of) subgroup differences through a priori designed RCTs. However, until further evidence is available, the exploratory findings from our post hoc subgroup analyses should not directly affect the current state of treatment for preschoolers with ADHD.

Finally, we also found that higher baseline ADHD symptom severity was associated with worse prognosis at endpoint, irrespective of the assigned treatment group. This finding is indicative that preschool children with more severe ADHD symptoms may still require additional interventions after trialing stimulants. Therefore, advancements in the treatment of ADHD in preschool children (eg, development of novel, or improvement of available, pharmacological and non-pharmacological interventions) may be particularly warranted for those who are more severely affected by this condition during this important developmental period.

Our study does have limitations. First, SPD489-347 had large discontinuation rates, which may introduce bias in their findings. Second, the MAPPA study did not collect information on race or ethnicity from their participants. Given that race is a social construct, we cannot determine whether, and how, our findings translate to different countries. Third, all RCTs included participants with moderate or severe baseline core ADHD symptoms, and we cannot determine how our findings translate to preschoolers with mild ADHD symptoms. Fourth, except for age (categorical) and sex, we had data for only 2 of the 3 RCTs for all putative moderator and predictor variables that were explored in this study. This inconsistency across RCTs from a relatively small field illustrates the wide variability in assessment instruments across studies, which in turn hinders the ability to draw conclusions from their data. We hope that our study underscores the importance of developing core measurement sets for RCTs in the field.[53] Fifth, our meta-analyses at times combined data from different stimulants (lisdexamfetamine, methylphenidate) given the small number of studies that were available, and we were unable to evaluate potential differences across medications. Finally, the approach combining subgroup data from PATS allowed adding data only regarding putative categorical variables. Because moderator analyses are exploratory in nature, it is possible that some of the categories were created to enhance discoverability in PATS. Indeed, we failed to replicate the only previous significant finding from PATS for the number of psychiatric comorbidities in MAPPA. Specifically, we showed that a higher number of baseline psychiatric comorbidities was associated with larger, and not smaller, reductions in ADHD symptom severity at endpoint with stimulants vs placebo. Relatedly, for race and ethnicity, the findings are also limited by the data that were collected and reported in PATS, as White, Black, and Hispanic participants were all grouped under the same category "ethnicity." Furthermore, we could not examine the moderating and predicting effect of individual comorbidities (eg, anxiety or conduct disorder), which would have been more informative clinically, as those data were not examined or reported in PATS.

In conclusion, this study collated moderator analyses from the 3 landmark randomized controlled trials of stimulants vs placebo for preschoolers with ADHD and showed that race, but not older age or baseline ADHD symptom severity, may moderate the efficacy of stimulants for preschool ADHD. Given the post hoc nature of subgroup analyses, these findings generate hypotheses that should be examined in future empirical studies.

CRediT authorship contribution statement

Luis C. Farhat: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Luisa Shiguemi Sugaya:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Michael H. Bloch:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Ann Childress:** Writing – review & editing, Visualization, Conceptualization. **Samuele Cortese:** Writing – original draft, Visualization, Conceptualization. **Daniel Fatori:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Giovanni A. Salum:** Writing – review & editing, Visualization. **Luis Augusto Rohde:** Writing – review & editing, Visualization, Conceptualization. **Guilherme V. Polanczyk:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Conceptualization.

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